

Densification behaviour of mixtures of hard and soft powders under pressure

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Received 1 April 1999; received in revised form 1 August 1999; accepted 7 December 1999

Abstract

The densification behaviour of mixtures of soft and hard powders during pressing is investigated. Several experimental studies using materials with miscellaneous mechanical and morphological features are summarised. The influence of mixture characteristics on the densification kinetics is discussed and several mechanisms explaining the influence of hard particles on the densification behaviour are proposed. Two main cases are distinguished. When the applied pressure is low, with regard to the yield stress of soft particles, the densification is mainly due to particle rearrangement and is favoured by a high fraction of hard particles. When soft particle deformation is the main densification mechanism, hard particles hinder the densification, with more or less significance depending on whether they are mostly isolated, are grouped in aggregates or form a percolating network. © 2000 Elsevier Science S.A. All rights reserved.

Keywords: Densification behaviour; Hard and soft powders; Pressing

1. Introduction

Various industrial processes involve the consolidation of mixtures of powders with different mechanical features. For example, powder processing is a classical route for the fabrication of metal–ceramic composite materials. The volume fraction of ceramic phase may vary from 10%, for example in structural components, to more than 50% for electrical or nuclear applications. “Soft” metal particles and “hard” ceramic powders are mixed together and possibly granulated. The mixture is next consolidated and sintered, either by two successive stages, die pressing at room temperature and natural sintering, or by one single operation, as hot isostatic pressing. The densification during cold or hot pressing is strongly influenced by the characteristics of the mixture, such as the phase fractions and the size, shape and strength of both types of particles. To control the consolidation process and optimise the composition of the mixture, it is thus important to antici-

pate the densification behaviour of such materials under pressure.

Numerous experimental studies have investigated the densification of hard and soft powder mixtures during die compression or isostatic pressing. These have been achieved either on industrial powders [1–7] or on model materials [8–14]. In most cases, the densification was mainly caused by the plastic deformation of soft particles. These experiments showed in particular that the densification was delayed when the fraction of hard particles was high, when the hard particles were small with respect to soft particles, and when they had a high aspect ratio (as fibres or plates) or an irregular shape. However, no general analysis of the mechanics of densification of such systems has been proposed so far. Indeed, extensive literature has also been devoted to the sintering of powder mixtures. Since sintering involves specific densification mechanisms with regard to compaction, it will not be considered here.

The present paper makes a survey of the behaviour of powder mixtures under pressure by reviewing experimental data obtained by the author and co-workers with miscellaneous materials in various conditions of compaction. In all examples, hard particles can be considered as non-

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deforming under applied pressures. The effects of volume fraction, size ratio, particle shape and strength on the densification behaviour are successively analysed. The physical phenomena responsible for the observed behaviour are next discussed with the support of several models of the literature. Emphasis is laid on the problem of mixtures with a high fraction of hard particles, which has not been much investigated so far, though it is of great practical interest. Unless otherwise specified, all materials have been prepared by mixing the powders with a Turbula shaker–mixer in dry conditions and have been directly poured into the die before compression.

The relative density is defined as the weight density of the porous material divided by the weight density of the fully dense material. An important parameter characterising a powder mixture is the size ratio, r , which is defined as the average size of soft particles divided by the average size of hard particles.

2. Influence of volume fraction and size ratio

The volume fraction of hard particles is indeed a parameter that significantly affects the compressibility of powder mixtures. This is clearly shown with the first example [1] that concerns a superalloy spherical powder with an average size of 50 μm , mixed with either 18 or 33 vol.% alumina particles with angular shape and an average size of 25 μm . The mixtures have been densified by hot isostatic pressing under 100 MPa at 1000°C. At that temperature, the superalloy particles are visco-plastic whereas the alumina particles are rigid. Hot-isostatic-pressing tests interrupted after various holding times allow estimating isothermal densification curves (Fig. 1). The negative influence of alumina particles is observed. For example, after pressing for 3 h, the relative density of the 35% alumina mixture is only 0.86, compared to 0.95 obtained with the 18% alumina mixture and 0.995 obtained with pure Astroloy powder. Typical microstructures of 35%

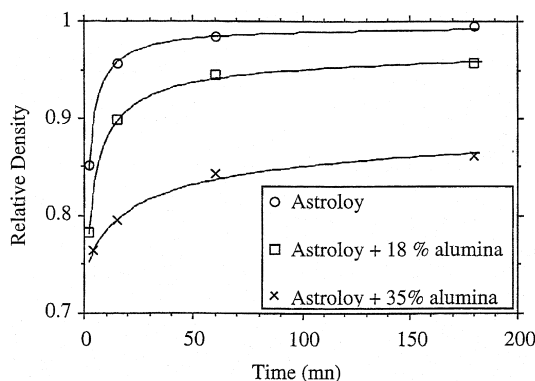


Fig. 1. Densification curves of pure Astroloy superalloy powder and mixtures of Astroloy powder and alumina particles, fitted from the results of hot isostatic pressing tests at 1000°C under 100 MPa [1].

alumina mixture after densification appear in Fig. 2. Fig. 2a shows that superalloy particles should be deformed to a large extent to fill the porosity located near the contacts of alumina particles between each other. A fine investigation of such micrographs by image analysis proved that the pores became preferentially located around inclusions as the densification progressed [15]. The break-up of alumina inclusions, which is observed in Fig. 2b, likely occurred at the early consolidation stage since some cracks are filled with superalloy. This suggests the existence of a continuous network of inclusions, which supports part of the applied pressure.

The effect of the size ratio, r , has been investigated with mixtures of titanium alloy powders and 22 vol.% titanium diboride particles. Titanium diboride particles had an average size of 10 μm . Several size ranges of titanium alloy powder had been chosen to prepare mixtures with various size ratios between 2 and 10. These mixtures have been consolidated by hot isostatic pressing at 800°C [3]. As in the previous example, metal particles were visco-plastic and spherical whereas ceramic inclusions were rigid and angular. The densification was faster when the size ratio was lower. For example, after pressing for 3 h at 800°C under 200 MPa, the relative density of the mixture with the lowest size ratio is 0.98, whereas the density of the mixture with the highest size ratio is 0.92. Fig. 3 compares the microstructure of both compacts. In the first case (Fig. 3a), the residual pores are located inside clusters formed by ceramic particles. In the second material (Fig. 3b), a continuous network of ceramic particles is observed and large pores still exist at the end of the test.

The combined effect of volume fraction and size ratio has been investigated by Hénon et al. [6], who densified mixtures of silver and tungsten carbide powders by die pressing at room temperature. The volume fraction of WC particles was between 20% and 40% and the size ratio between 0.1 and 10. Silver particles had irregular shapes and underwent plastic deformation during compression. WC particles were angular and rigid. To investigate a large range of size ratios, two silver powders with average sizes of 5 and 15 μm , respectively noted Ag_m and Ag_g were used. Fig. 4 shows the relative density obtained with pure silver powders and with all mixtures prepared with either powder after die compression to 600 MPa. The mixtures with a size ratio of 2.5 or 10 were prepared with Ag_g powder and the others with Ag_m powder. Notice that the two silver powders, which likely have different microstructural features, did not reach the same density. As expected, the density increases with decreasing fraction of ceramic particles and with decreasing size ratio. For example, the same density, 0.85, has been reached with the mixture containing 40 vol.% of small WC particles ($r = 2.5$) and with the mixture containing 25 vol.% of very small WC particles ($r = 10$). When WC particles are much larger than Ag particles ($r \ll 1$), they do not significantly hinder the densification, even with a volume fraction of 40%.

3. Influence of the shape of hard particles

In all powder mixtures presented above, the hard inclusions had an angular shape due to the fabrication technique. To investigate the influence of inclusion morphology, visco-plastic lead beads of diameter 500 μm have been mixed with rigid inclusions of different sizes and morphologies and compacted either under isotropic pres-

sure or in a rigid die at 150°C [14]. The obtained data show in particular that angular inclusions have a more drastic effect on the densification kinetics than spherical ones. As an illustration, densification curves drawn from isostatic pressing tests for mixtures with a size ratio of 1 are presented in Fig. 5. This result can be understood from Fig. 6, which shows micrographs of compacts with a relative density of 0.9. It appears that lead particles should

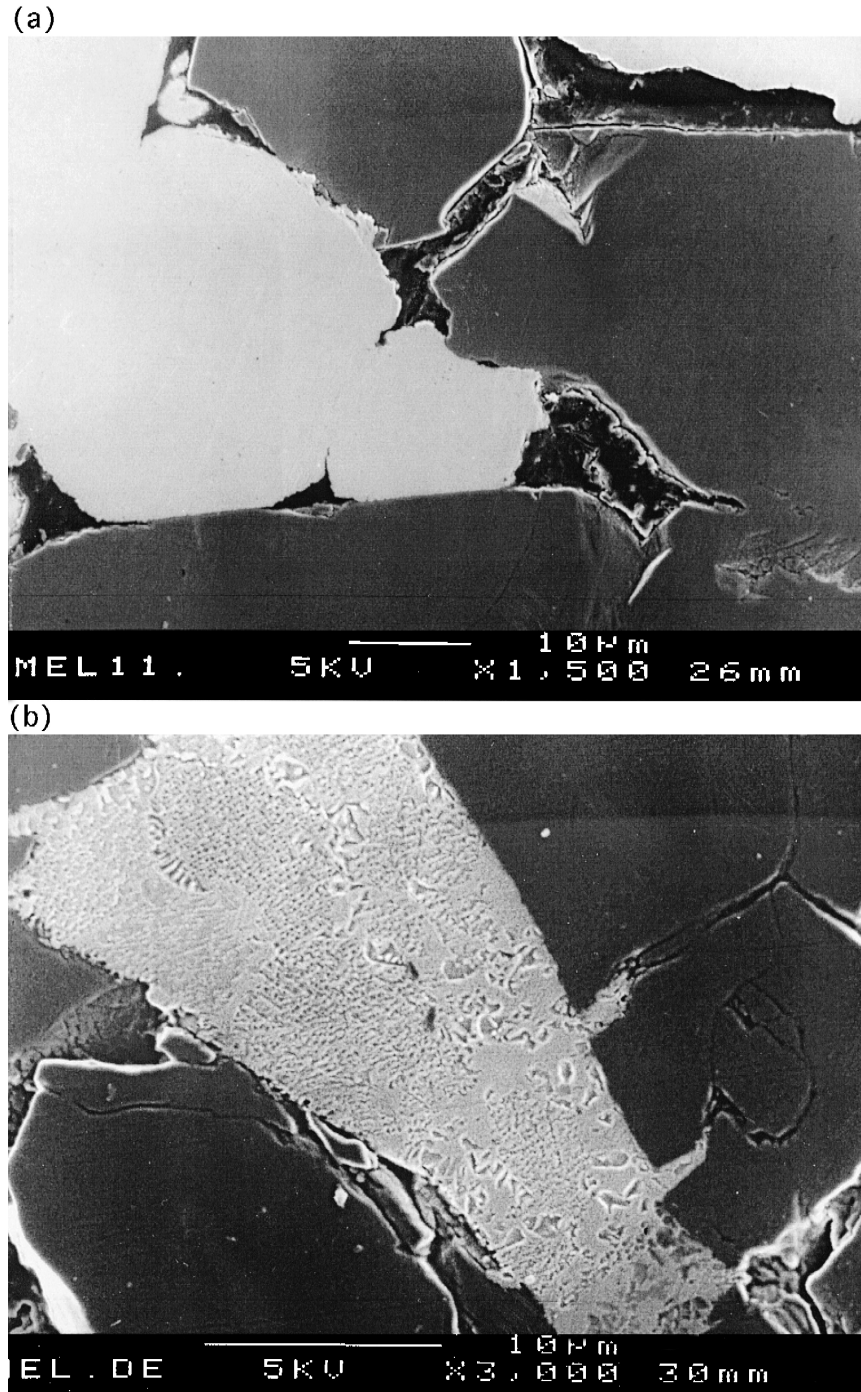


Fig. 2. Micrographs of mixtures of superalloy powder (clear) and 35% volume fraction of alumina particles (dark) after 3-h pressing at 1000°C under 100 MPa [1].

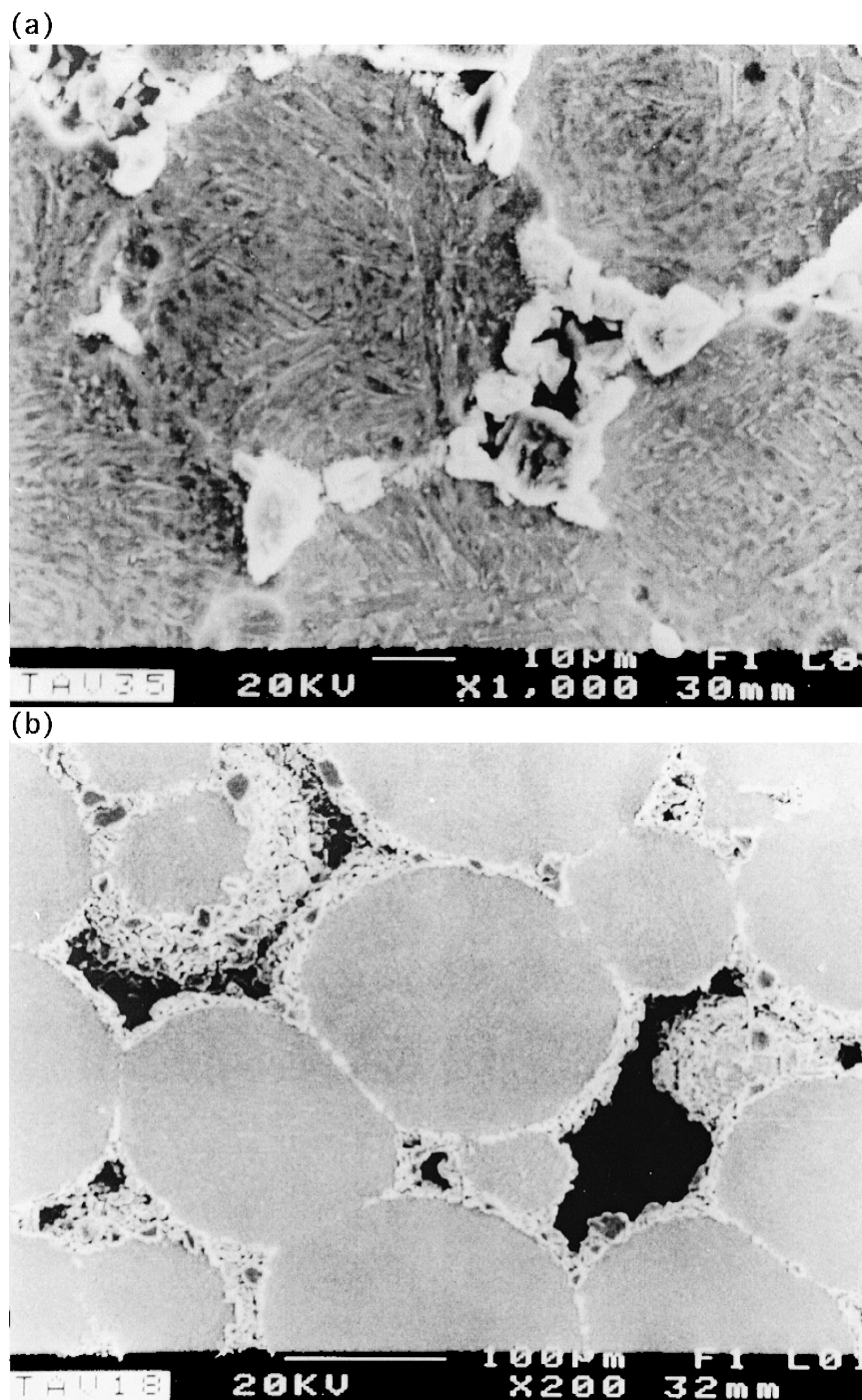


Fig. 3. Micrographs of two mixtures of titanium alloy powder (dark) and 22% volume fraction of titanium diboride particles (clear) with different size ratios after 3-h hot isostatic pressing at 800°C under 200 MPa [3].

be less heavily deformed to surround inclusions or fill the gaps between inclusions when the inclusions are spherical (Fig. 6a) than when they are irregular (Fig. 6b).

4. Influence of mechanical strength of soft particles

A mixture of aluminium powder smaller than 10 μm and 60 vol.% carbide particles of average size 60 μm has

been compacted in a closed die at room temperature and at 450°C. The resulting densification curves are compared in Fig. 7. The densification is easier when the temperature gets higher. After compression to 600 MPa at room temperature, the relative density was about 0.85. At 450°C, the same mixture densified up to 0.91 of relative density under the same pressure. Reaching such a high density with 60 vol.% hard particles is surprising. In the initial packing

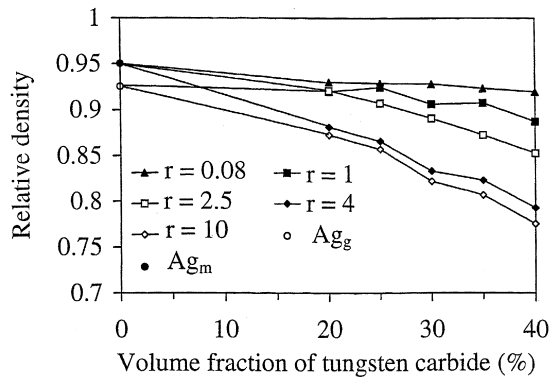


Fig. 4. Relative density of pure Ag powders and Ag–WC powder mixtures with various size ratios (r) after die compression at 600 MPa [6]. Ag_m and Ag_g refer to silver powders with average sizes of 5 and 15 μm , respectively.

(relative density ≈ 0.5), carbide particles fill about 30% of the total volume and, thus, already form a continuous skeleton. In 91% dense compact, they fill 55% of the volume, which is close to their maximal packing density. As no carbide particle breaking or deformation has been observed, important rearrangement of carbide particle skeleton certainly occurred during densification. This rearrangement is controlled by the deformation of aluminium particles, which is easier at high temperature because the yield stress of aluminium decreases with increasing temperature.

The last example concerns mixtures of tungsten carbide and cobalt particles as used for the production of cemented carbide [5]. Composite granules have been prepared with the following procedure. The powders have first been mixed and ground in a ball mill with acetone and 2% weight fraction of polyethylenglycol, which is used as the binding agent for granulation. The resulting mix has next been dried in a rotating drier. Granules with an average size of 100 μm are then obtained. Tungsten carbide particles have an average size of 2 μm and cobalt particles are sub-micronic. Various mixtures, with a volume fraction of cobalt between 0% and 24%, have been compacted in a closed die at room temperature. As observed in Fig. 8, the densification is improved when the fraction of cobalt is lower. This outcome is reversed to the results of the preceding studies. The reason is simple: in the investigated range of compacting pressure, densification mainly occurs by particle rearrangement and not by plastic deformation as previously. The friction between “hard” carbide particles is likely lower than the friction between “soft” cobalt particles or between a cobalt particle and a carbide particle. Hence, the rearrangement is easier when the material contains a lower fraction of cobalt particles.

5. Summary of densification mechanisms

The densification mechanisms of a uniform powder during compression are well known. At low pressure, the

densification results from particle rearrangement. The increase in density is particularly important when the initial density of the powder is low. Ideally, pure rearrangement should lead to the maximum packing density of the powder, which depends on the size distribution and on the shape of the particles. The continuation of the densification can result either from the plastic- or viscous-deformation of the particles or from their fragmentation, depending on whether these particles are ductile or brittle at the temperature of the compression. Particle deformation leads to complete densification if the pressure is high enough or if the holding time is long enough. Fragmentation allows compact packing but cannot provide a fully dense material.

Now, let us turn to the problem of mixture of soft and hard powders. First, we will briefly consider the case when the compaction pressure is not high enough to allow the deformation of soft particles. This situation has been discussed above when we analysed cobalt–carbide mixtures. Since ceramic particles were harder and more brittle than metal particles, the densification by rearrangement or fragmentation was favoured when the fraction of ceramic particles was high. This result can be generalised. At low density, a mixture with high volume fraction of hard particle densifies more easily because of the lower coefficient of friction at the interface between hard particles.

Let us assume now that the densification mainly results from the deformation of soft particles and that hard particles are much stronger than soft ones. Therefore, all experimental data show that the densification is easier when the fraction of hard particles is lower. We will distinguish three cases.

5.1. Isolated inclusions

When the fraction of hard particles is low, most of these particles are isolated from each other. As these are rigid and replace soft particles, they inevitably hinder the densification with regard to homogeneous metal powder. However, if pressure is high enough, or time long enough, soft

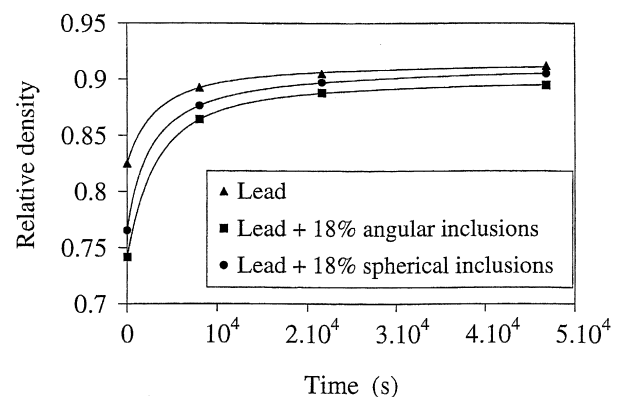


Fig. 5. Densification curves of lead powder and mixtures of lead powder and 18% volume fraction of either spherical or angular inclusions, fitted from the results of isostatic pressing tests at 150°C under 8 MPa [14].

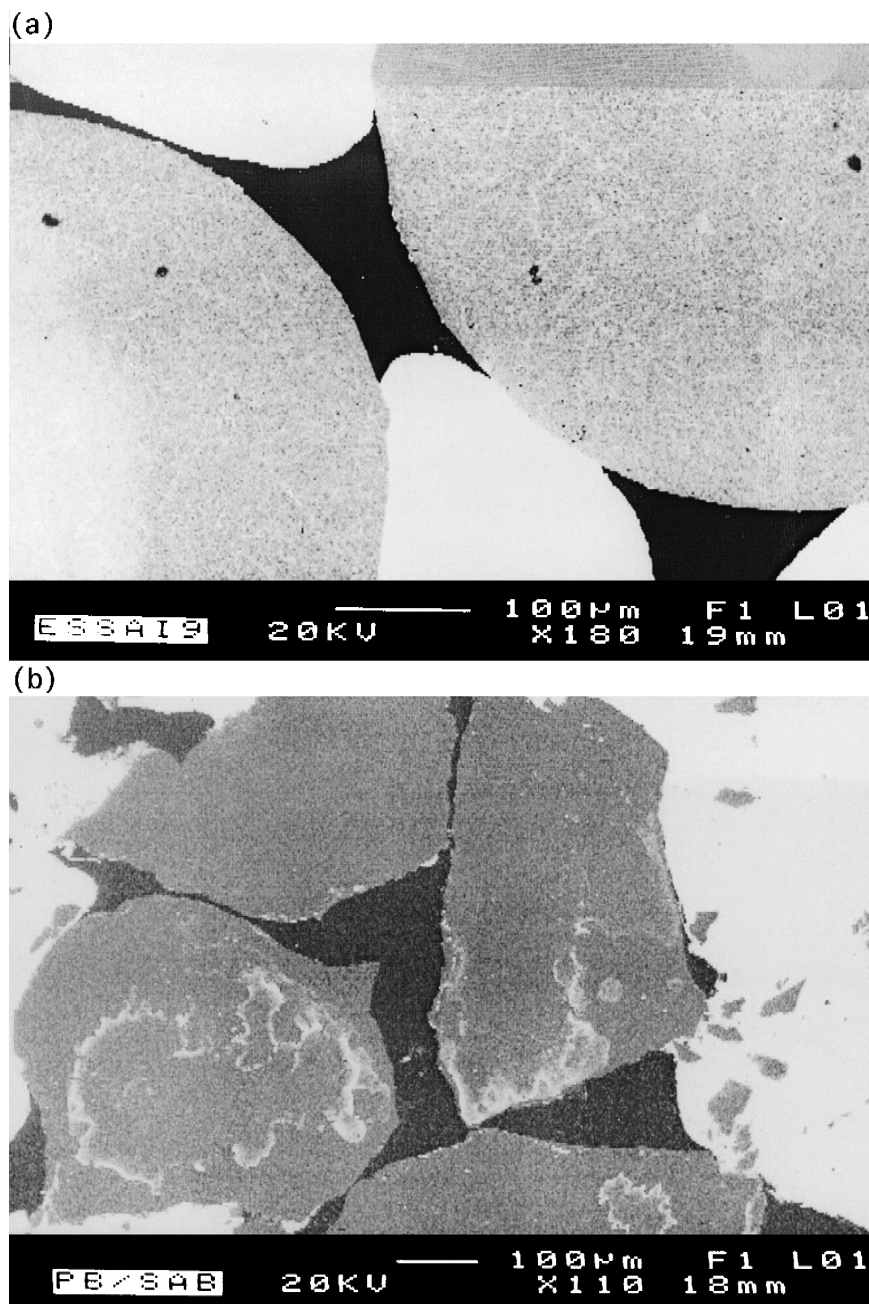


Fig. 6. Micrographs of mixtures of lead particles (clear) with either spherical or angular inclusions (dark) at a relative density of 0.9 [14].

particles can deform until all pores are filled. The deformation required to fill the porosity is higher when hard particles have a high specific area, i.e., when they are small or have an irregular shape. When hard particles are much larger than soft particles, the densification rate is almost unaffected by the inclusions, as observed for Ag–WC mixtures (Fig. 4).

The effect of inclusion morphology has been modeled by Delie and Bouvard [14], who calculated with a numerical method the deformation of square arrays of circular visco-elastic particles and hard inclusions with miscella-

neous shapes under isotropic pressure. These calculations confirmed that inclusions with high surface area and angularities were unfavourable to the densification because of the strong deformation of the soft phase required to reach high densities.

5.2. Aggregation

Aggregates form when the fraction of hard particles increases, even when the mixture is homogeneous. Filling the narrow porosity located around interparticle contacts,

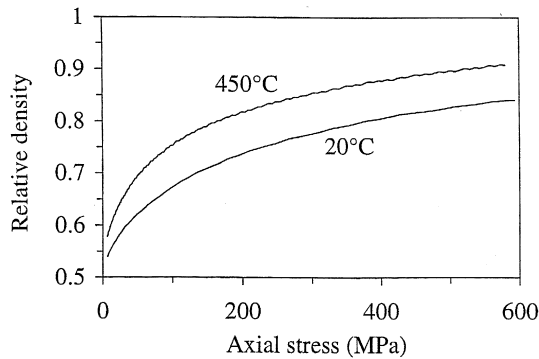


Fig. 7. Relative density vs. axial stress curves of a mixture of aluminium powder with 60% volume fraction of carbide particles during die pressing at two temperatures, 20°C and 450°C.

which has been called “excluded volume” by Lange et al. [9], requires heavy deformation of the soft particles, i.e., high compacting pressure or long creep time. The required deformation is heavier when hard particles are smaller. Pores located inside large aggregates can even be inaccessible to the soft phase. In this case, it is not possible to reach full densification. This situation has been met for example in Figs. 3a and 6b.

5.3. Percolation

Lastly, we consider the case when the fraction of hard particle is still higher, so that a percolating particle network forms. This network — or “skeleton” — supports part of the external pressure and, thus, relieves the metal particles, partially or totally. The densification of the mixture by plastic deformation of only soft particles is then difficult, or even impossible. Deformation of the skeleton of hard particles is then required. Such deformation can result from rearrangement, plastic deformation or fragmentation of the particles of the network according to the mechanical properties of these particles and the compactness of the network. Rearrangement of the network is the most usual mechanism. It is necessarily controlled by the deformation of the soft phase surrounding this network.

The example of Fig. 5 clearly illustrates this point; the skeleton of carbide particles is more easily rearranged when the temperature gets higher, i.e., when the applied pressure is lower with regard to the yield stress of aluminium. In superalloy–alumina composite presented above, the rearrangement of alumina network is likely more difficult due to the irregular shape of the particles. The densification is almost stopped, although the breaking of brittle alumina particles allows a little rearrangement of the skeleton. The percolation of ceramic particles has, thus, a dramatic effect on the densification of superalloy–alumina mixture, whereas it does not abruptly affect the behaviour of aluminium–carbide mixture.

This analysis is supported by the results obtained by Jagota and Scherer [16] from numerical modeling. These authors calculated the effective viscosity of mixtures of about 1000 spheres of the same size with either very high or very low viscosity. They found that the mechanical percolation threshold, i.e., the limit fraction of hard particles above which the viscosity gets high, strongly depended on interparticle contact law. If bonded contacts are assumed, the mechanical threshold is exactly equal to the geometrical threshold, 0.32. If sliding is allowed, the mechanical threshold is about 0.96, which means that the mixture can be soft even with a fraction of hard particles much above the geometrical percolation threshold. This result confirms that the densification behaviour of a powder mixture with a high fraction of hard particles depends mainly on the capability of rearrangement of the hard particle skeleton.

5.4. Critical volume fractions

To anticipate the densification behaviour of a powder mixture, one should know in which situation this mixture happens to be among the three situations that have just been presented; isolated inclusions, aggregates, percolation. The volume fractions of hard particles corresponding to the thresholds between these three cases strongly depend on the size ratio. These critical fractions are lower when the hard inclusions are smaller than soft particles, as schematised in Fig. 9. A low volume fraction of small particles can form aggregates and even percolate. This property appears for example in the micrographs of Fig. 3; 22% of TiB_2 particles form a compact network only when these particles are much smaller than the titanium particles.

The numerical simulation of particle packing can be helpful to estimate the critical volume fractions. To illustrate this, we present results obtained by Bouvard and

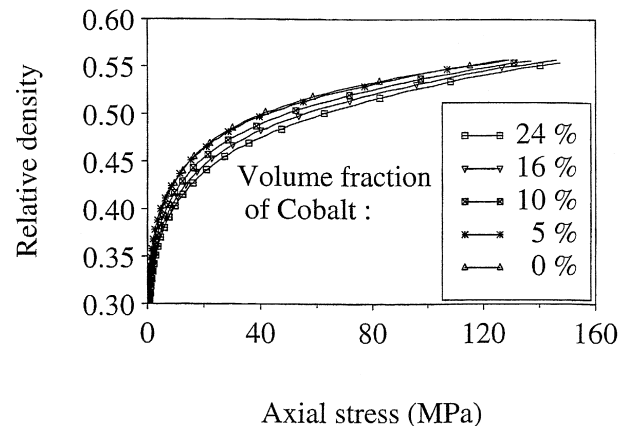


Fig. 8. Relative density vs. axial stress curves of WC–Co mixtures with various volume fractions of Cobalt powder during die pressing at room temperature [5].

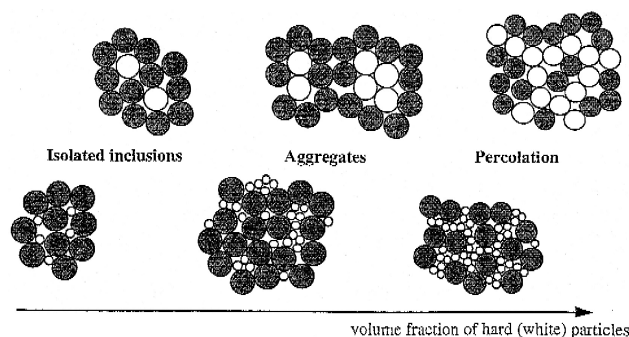


Fig. 9. Schematic microstructure of powder mixtures according to particle size ratio and phase fractions.

Lange [17], who modeled packing of about 10 000 mono-sized spheres into a cubic box. Each sphere was randomly considered as an “inclusion” or a “matrix particle”, so that a prescribed volume fraction of inclusions was obtained. All the aggregates formed by the inclusions in the packing were identified. Fig. 10 presents the fraction of inclusions that are inside a cluster of a given size. We deduce for example that in a mixture with a fraction of inclusions of 0.1, 58% of these inclusions are isolated (i.e., in a cluster of size 1) and 23% in a pair. With 20% of inclusions, only 32% of the inclusions are isolated, 18% are in a pair, etc. The first percolating network appears for a fraction of inclusions of 0.35. When the fraction of inclusions is 0.5, 95% of the inclusions are in a single percolating network. It can be deduced, for example, that in lead-based mixtures presented in Figs. 5 and 6, most inclusions are isolated or in small aggregates.

The volume fraction of inclusions corresponding to the percolation threshold has been drawn by the same authors from similar simulations [17]. It is plotted in Fig. 11 as function of matrix particle to inclusion size ratio. This critical volume fraction is 0.32 for mono-sized mixture and decreases with increasing size ratio. For example, it is equal to 0.18 for a size ratio of 2. This result suggests for

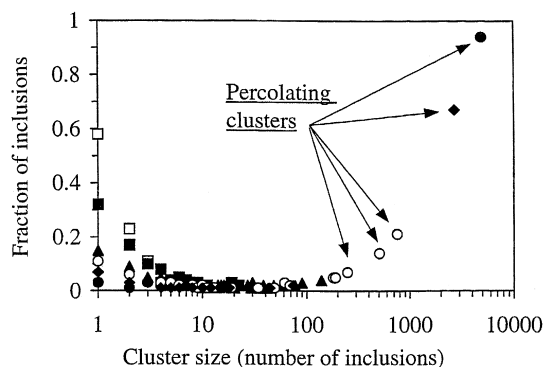


Fig. 10. Clustering in mixtures of monosized spheres with different fractions of inclusions: 0.1 ($\square$), 0.2 ($\blacksquare$), 0.3 ($\blacktriangle$), 0.35 ($\circ$), 0.4 ($\blacklozenge$), 0.5 ($\bullet$) (from Ref. [17]).

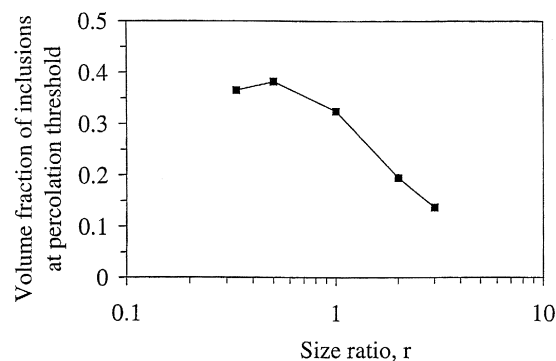


Fig. 11. Volume fraction of inclusions at percolation threshold as function of size ratio, obtained from numerical simulation of mixtures of spheres [17].

example that in the 18% alumina mixture presented above, alumina particles hardly form a percolating network. Therefore, this network should be loose and not very prejudicial to the densification.

6. Conclusion

The major phenomena explaining the densification behaviour of mixtures of hard and soft particles under isotropic or quasi-isotropic pressure are rather well understood, thanks to experimental data obtained for mixtures with miscellaneous mechanical and morphological features. Two main cases have been distinguished. When the yield stress of soft particles is too high with regard to the applied pressure, the densification progresses by particle rearrangement, and is most often favoured by a high fraction of hard particles. When soft particle deformation is the main densification mechanism, hard particles hinder the densification, above all when they are smaller than soft particles. Three situations may occur depending on the volume fraction of hard particles and on the size ratio. Hard particles may be mostly isolated from each other, may form aggregates, or create a percolating network. The third situation is indeed the most critical. The densification can only proceed by the deformation of the network, as a consequence of either rearrangement, plastic deformation or fragmentation of the hard particles. If such deformation is not possible under applied pressure, the densification is stopped.

Acknowledgements

The author is grateful to his numerous co-workers, J.M. Chaix, F. Delie, A. Hénon, M. Lafer, F.F. Lange, C. Martin, P. Thibault and S. Roure, who contributed to obtain and interpret the data presented in this paper.

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